where $\mathbf{n} \cdot \hat{\boldsymbol{\sigma}} = n_x \hat{\sigma}_x + n_y \hat{\sigma}_y + n_z \hat{\sigma}_z$. This operator corresponds to a rotation by $\phi$ around an axis pointing along $\mathbf{n}$. Similarly to above, we also have

$$\hat{R}_{\mathbf{n}}(\phi) = \cos\left(\frac{\phi}{2}\right)\mathbb{1} - i\sin\left(\frac{\phi}{2}\right)\mathbf{n} \cdot \hat{\boldsymbol{\sigma}}. \tag{2.56}$$

## 2.3 Entanglement and density operators

As soon as we consider composite systems – systems made up of two or more parts – we open up for correlations among the subsystems. Quantum systems can be correlated in new ways that classical systems cannot. It is also very often the case that a system of interest is actually a subsystem of some larger system which we are not interested in or unable to describe in detail. For example, a qubit implemented in some physical system, say an atom, cannot generally be perfectly isolated from the environment. Therefore, we need ways to describe states, evolution, and measurements of parts of systems. The axioms in Sec. 2.1 are give a perfectly valid and complete formulation of quantum mechanics, but they assume a closed system. In a sense, they describe the entire universe. When considering only a small part of the universe, i.e. a subsystem, they need to be revised. In particular, in general for subsystems, states are *not* rays and evolution is *not* unitary.

### 2.3.1 Entanglement for pure states

We start by considering quantum correlations when the total system is closed (so the axioms from before apply and its state is still a ray, i.e. a pure state). As explained in Sec. 2.1.6, the state space of a bipartite system is a product $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$. If the states of subsystems A and B are independent, then the state of the total system is also a product

$$|\psi\rangle_{AB} = |\psi_A\rangle_A \otimes |\psi_B\rangle_B. \tag{2.57}$$

Not all bipartite states (2.32) can be factorised like this. The ones that do not factorise necessarily contain some quantum correlations. Any state of a bipartite system which can be written on this product form, for some choice of states $|\psi_A\rangle_A$ on A and $|\psi_B\rangle_B$ on B, is said to be *separable*. Any state which is not separable is said to be *entangled*.

### 2.3.2 Schmidt decomposition and Schmidt number

For any bipartite pure state $|\psi\rangle_{AB}$, it is always possible to find orthonormal bases $\{|n\rangle_A\}$ and $\{|n'\rangle_B\}$ for A and B, respectively, such that the state can be written

$$|\psi\rangle_{AB} = \sum_n \sqrt{p_n}|n\rangle_A|n'\rangle_B, \tag{2.58}$$

with real, nonnegative coefficients, $p_n \geq 0$. This is known as the *Schmidt decomposition* or *Schimdt form*.

The *Schmidt number* of a state is the number of nonzero coefficients in the Schmidt decomposition. All separable pure states are product states and have a Schmidt number of 1. A Schmidt number larger than 1 implies that the state is entangled.

where $\|\hat{A}\|_1 = \mathrm{Tr}\sqrt{\hat{A}^\dagger \hat{A}}$ is the trace norm, and the $\lambda_k$ are the eigenvalues of $\hat{\rho} - \hat{\sigma}$ (which are real since $\hat{\rho} - \hat{\sigma}$ is Hermitian). The trace distance is a quantum generalisation of the statistical distance (or total variation distance) between probability distributions (see Chap. 8). Like the fidelity, it is positive and bounded by unity

$$0 \leq T(\hat{\rho}, \hat{\sigma}) \leq 1, \tag{2.92}$$

When both states are pure, the trace distance and fidelity are equivalent in the sense that

$$T(|\phi\rangle, |\psi\rangle) = \sqrt{1 - |\langle\phi|\psi\rangle|^2} = \sqrt{1 - F}, \tag{2.93}$$

where $T(|\phi\rangle, |\psi\rangle)$ is understood as taken for the corresponding density operators. For general states, they are related through the inequalities

$$1 - \sqrt{F(\hat{\rho}, \hat{\sigma})} \leq T(\hat{\rho}, \hat{\sigma}) \leq \sqrt{1 - F(\hat{\rho}, \hat{\sigma})}, \tag{2.94}$$

with $T(\hat{\rho}, \hat{\sigma}) = 0$ if and only if $\hat{\rho}$ and $\hat{\sigma}$ are the same state, $\hat{\rho} = \hat{\sigma}$, and $T(\hat{\rho}, \hat{\sigma}) = 1$ if and only if $\hat{\rho}$ and $\hat{\sigma}$ have orthogonal supports.

The trace distance can be given physical meaning by relating it to the probabiity of distinguishing the states $\hat{\sigma}$ and $\hat{\rho}$. Given a source that prepares $\hat{\sigma}$ or $\hat{\rho}$ with equal probability, consider the problem of identifying the state with a binary measurement (i.e. the measurement must always provide a conclusive outcome, unlike in Example 2.1), while making as few errors as possible. This is known as minimum-error state discrimination (see Prob. 2.12 for the case of pure states). The minimum average error probability with an optimal, binary POVM, is

$$p_e = \frac{1}{2}(1 - T(\hat{\rho}, \hat{\sigma})). \tag{2.95}$$

For $T(\hat{\rho}, \hat{\sigma}) = 1$, the states are perfectly distinguishable, $p_e = 0$, while for $T(\hat{\rho}, \hat{\sigma}) = 0$ it is not possible to do better than random guessing, $p_e = \frac{1}{2}$.

### 2.3.7 Entanglement for mixed states

We have seen that a pure state of a bipartite system is entangled if it is not a product state. When extending the definition to mixed states, we need to take into account that classical mixtures of product states are also only classically correlated. A bipartite mixed state $\hat{\rho}_{AB}$ is *separable* if it can be expressed as a mixture of product states, i.e.

$$\hat{\rho}_{AB} = \sum_k p_k \, \hat{\rho}_{A,k} \otimes \hat{\rho}_{B,k}, \tag{2.96}$$

for some sets of states $\{\hat{\rho}_{A,k}\}$ and $\{\hat{\rho}_{B,k}\}$ on subsystems A and B, respectively, and some distribution $p_k \geq 0$, $\sum_k p_k = 1$.

#### Entanglement decision problem

In general, deciding whether a given density operator on a bipartite system is entangled is a difficult problem. Computationally, it is known to be NP-hard with respect to the system dimension. A complete characterisation of the set of separable states is known only in particular cases.